Day 6 Task Report

Title: Stateful Chatbot with Conversation Memory

TASK-06: Implementing a Stateful Chat Interface with Context Windows

Objective

The objective of this task is to build a functional CLI (Command Line Interface) chatbot using a live Large Language Model (LLM).Unlike previous tasks, this chatbot maintains conversation memory and responds based on previous interactions.The system simulates a Technical Recruiter, asking and evaluating interview-style questions.

Tools Used

- VS Code
- Git & GitHub
- MS Word

Libraries Used

- **google-genai:** To interact with Gemini API
- **dotenv:** To securely store API key
- **os:** To access environment variables

Concepts Used

1. API Integration

The chatbot connects to Gemini LLM using:

client = genai.Client(api_key=...)

2. Stateful Conversation (Memory)

A list is used to store conversation:

conversation_memory = []

Each user input and bot response is appended to this list.

3. Context Management

The full conversation history is sent to the model:

context = SYSTEM_PROMPT + "\n" + "\n".join(conversation_memory)

This helps the model understand previous messages.

4. Sliding Window (Buffer Limit)

To optimize performance and reduce token usage:

conversation_memory = conversation_memory[-10:]

Only the last 5 exchanges (10 messages) are kept.

5. System Persona

The chatbot is designed to behave as a Technical Recruiter:

"You are a professional Technical Recruiter..."

It never breaks character and always responds professionally.

Code Explanation

The program works as follows:

1. User enters input via CLI
2. Input is added to memory
3. Memory is trimmed using sliding window

4. Full context is sent to Gemini API
5. Model generates response
6. Response is added back to memory
7. Output is displayed

Errors Faced & Solutions

1. Model Not Found Error

The Wrong model name used
Solution: Updated to:

gemini-2.5-flash

2. Chatbot was not behaving like a recruiter

In the beginning, the bot was giving general answers instead of acting like a recruiter.

Solution
I fixed this by adding a **system prompt** that clearly defines its role as a Technical Recruiter.

3. Conversation context was not structured

Earlier, messages were not properly combined before sending to API.

Solution
I fixed this by creating a proper context using:

```
context = SYSTEM_PROMPT + "\n" + "\n".join(conversation_memory)
```

Output

```
👔💼 Technical Recruiter Chatbot (Gemini) - Type 'exit' to quit

You: Heyy
To get started, could you please state your full name and confirm the role you're interviewing for? Once we've done that, we can dive right in. Are you ready?
You: Yeah Sure. I'm Kajal Sankapal and I'm interested in Machine Learning Engineer role
Recruiter: Excellent, Kajal. Thank you for confirming. It's a pleasure to speak with you today about the Machine Learning Engineer role.

To begin, I'd love to hear a bit more about your background and experience. Could you please walk me through your resume, highlighting the projects or experiences that you feel are most relevant to thi
s Machine Learning Engineer position? Take your time.
You: Sure! I'm an AIML student with hands-on experience in building real-world projects like an IoT-based pipeline leakage detection system and LLM-based chatbots. I've also worked on data visualizatio
n using Power BI and I'm currently doing a Geodata Python course from ISRO, which strengthens my machine learning and analytical skills relevant to this role.
Recruiter: Thank you for that excellent overview, Kajal. It's clear you've been actively engaged in some very relevant areas, especially with the real-world projects and your continuous learning.

Let's dive a bit deeper into some of these. The **IoT-based pipeline leakage detection system** sounds particularly interesting and impactful. Could you elaborate on that project? Specifically, I'd lik
e to understand:

*   What was your role and what were your key contributions to this system?
*   What kind of sensor data were you working with, and what machine learning models or techniques did you employ to detect leakages?
*   What challenges did you face in building this system, particularly concerning data processing, model selection, or deployment, and how did you overcome them?
You: exit
Good luck with your career! 🚀
(.venv) PS D:\Qualtech\Daily Tasks\Day6>
```